

Polar Equations and Graphs



Recognizing polar graphs

1 Identify the graph of each polar equation:

a $r = 3$

b $\theta = \frac{\pi}{6}$

c $r = 2 + 2\cos \theta$

d $r = 4\sin 3\theta$

2 For the limaçon $r = 1 + 2\cos \theta$:

a Explain how you know it has an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole.

3 For the rose $r = 5\sin 2\theta$:

a State the number of petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs.

4 Determine whether the graph of $r = 2 + \sin \theta$ passes through the pole, and state the type of limaçon.

Symmetry tests

5 Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).

6 Test $r = 2 + 2\sin \theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).

Graphing from a table of values

7 For the cardioid $r = 1 + \cos \theta$, complete the table of r -values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.

a $r(0)$

b $r(\frac{\pi}{2})$

c $r(\pi)$

d $r(\frac{3\pi}{2})$

8 For the circle $r = 4\sin \theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.

Rose curves in depth

- 9 For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).
- 10 Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.